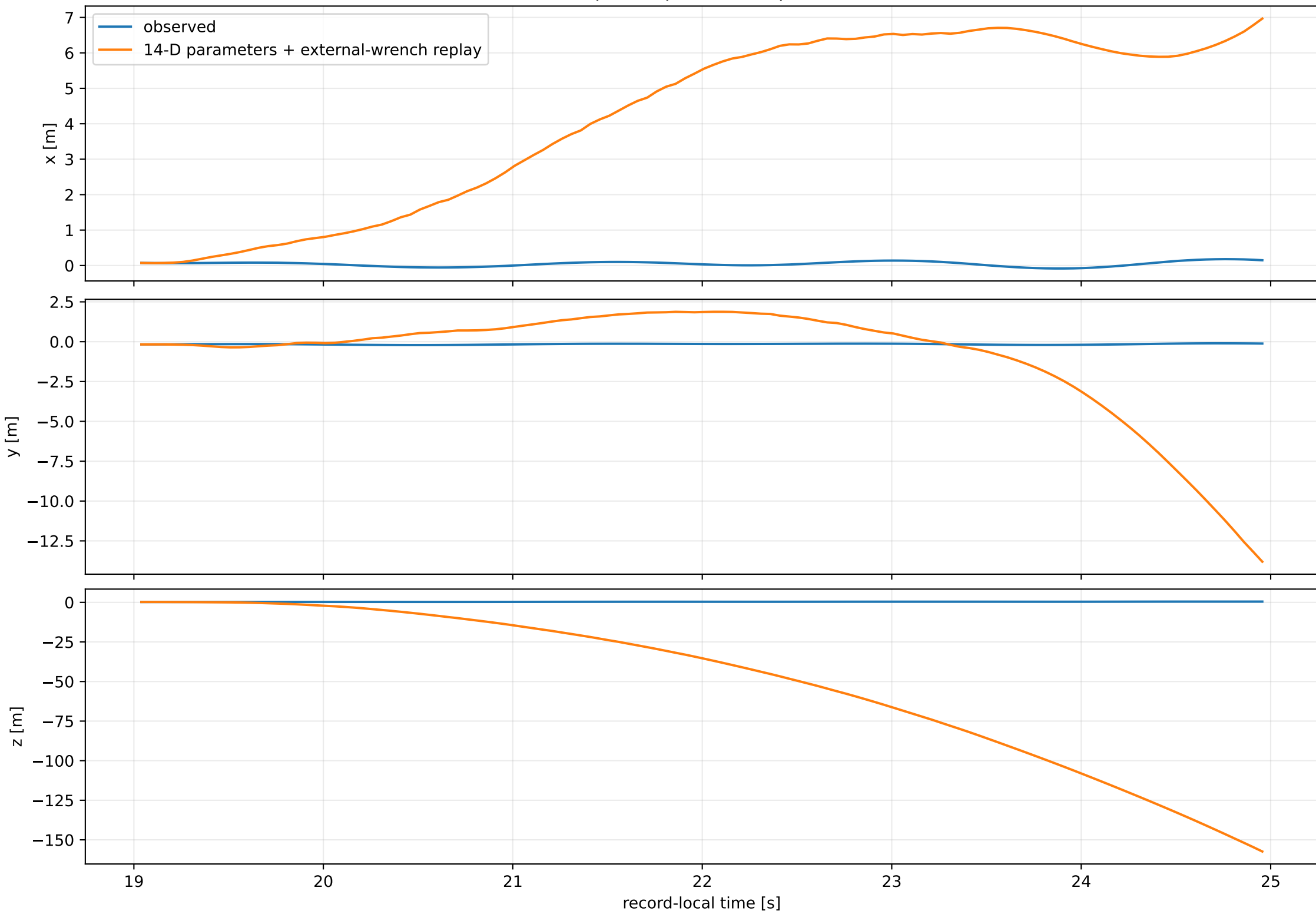
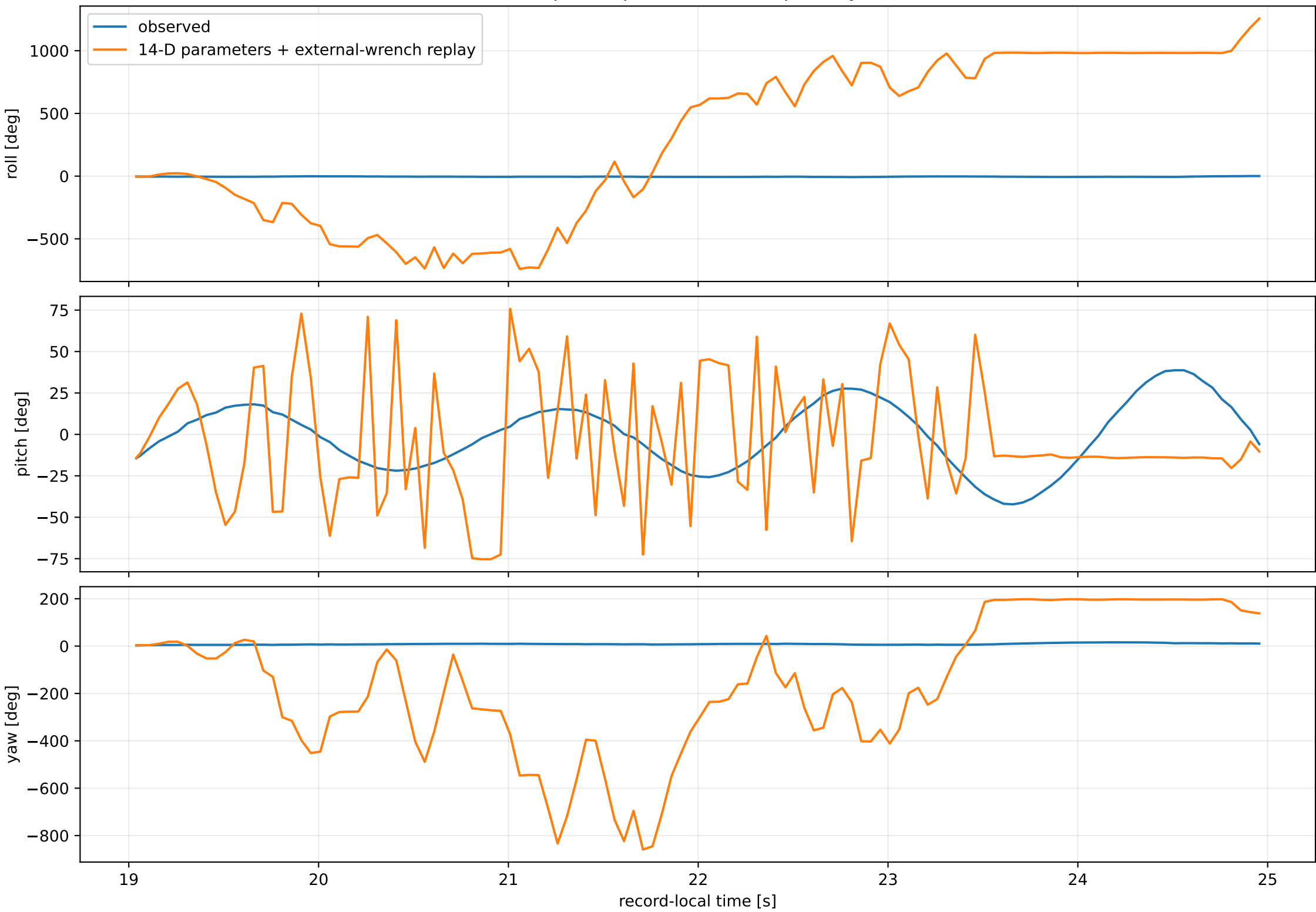


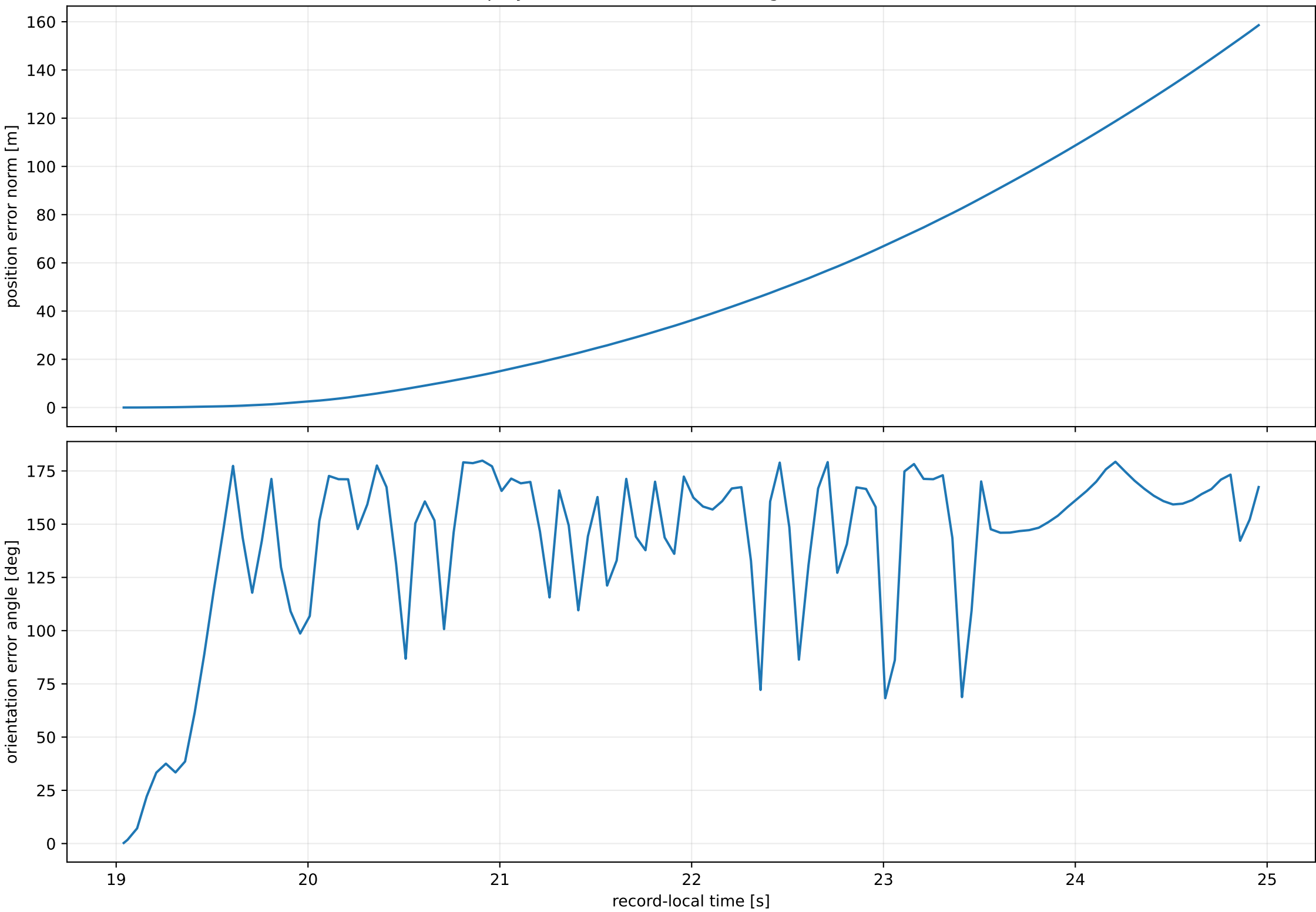
Fine pose reproduction: position



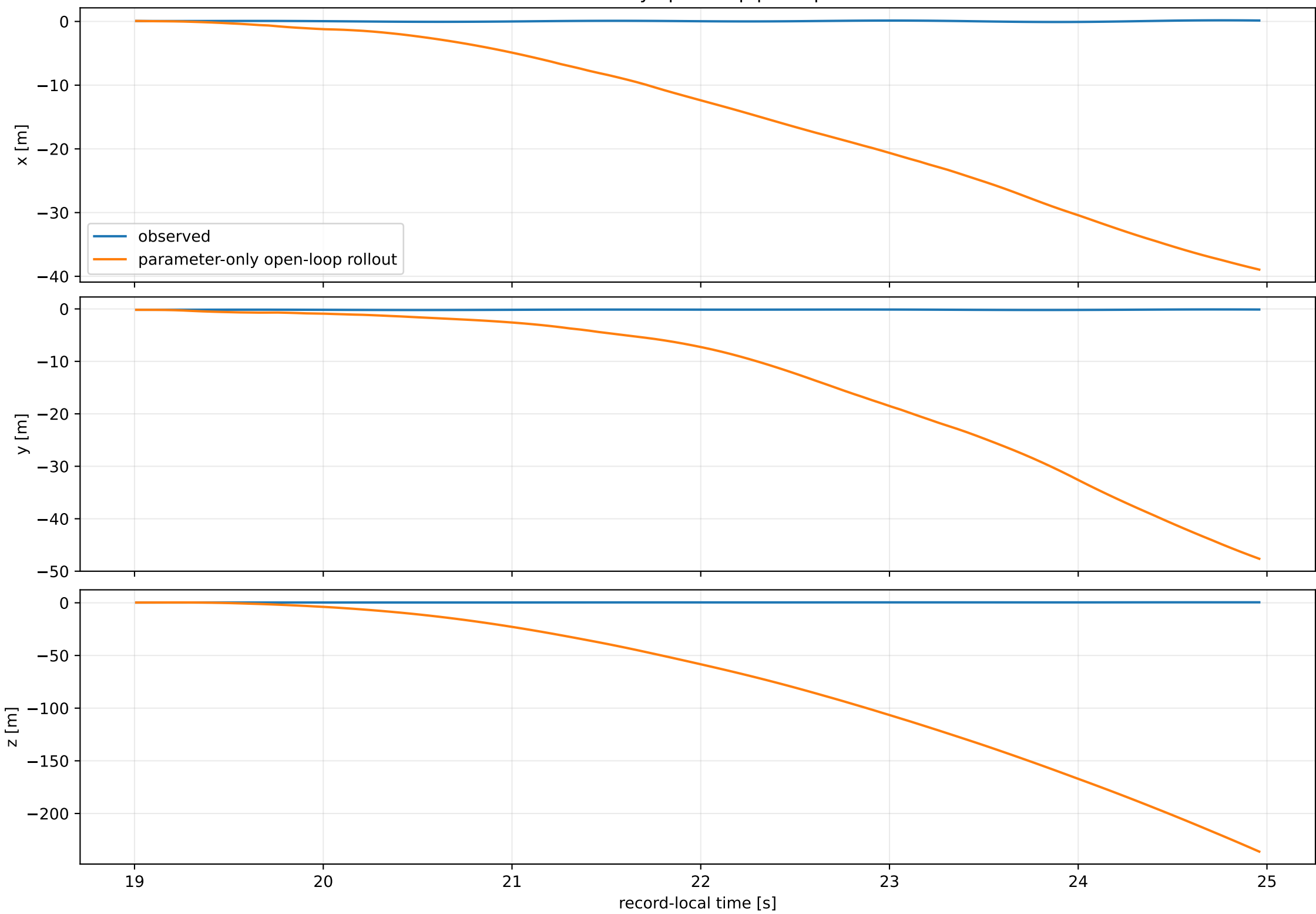
Fine pose reproduction: roll / pitch / yaw



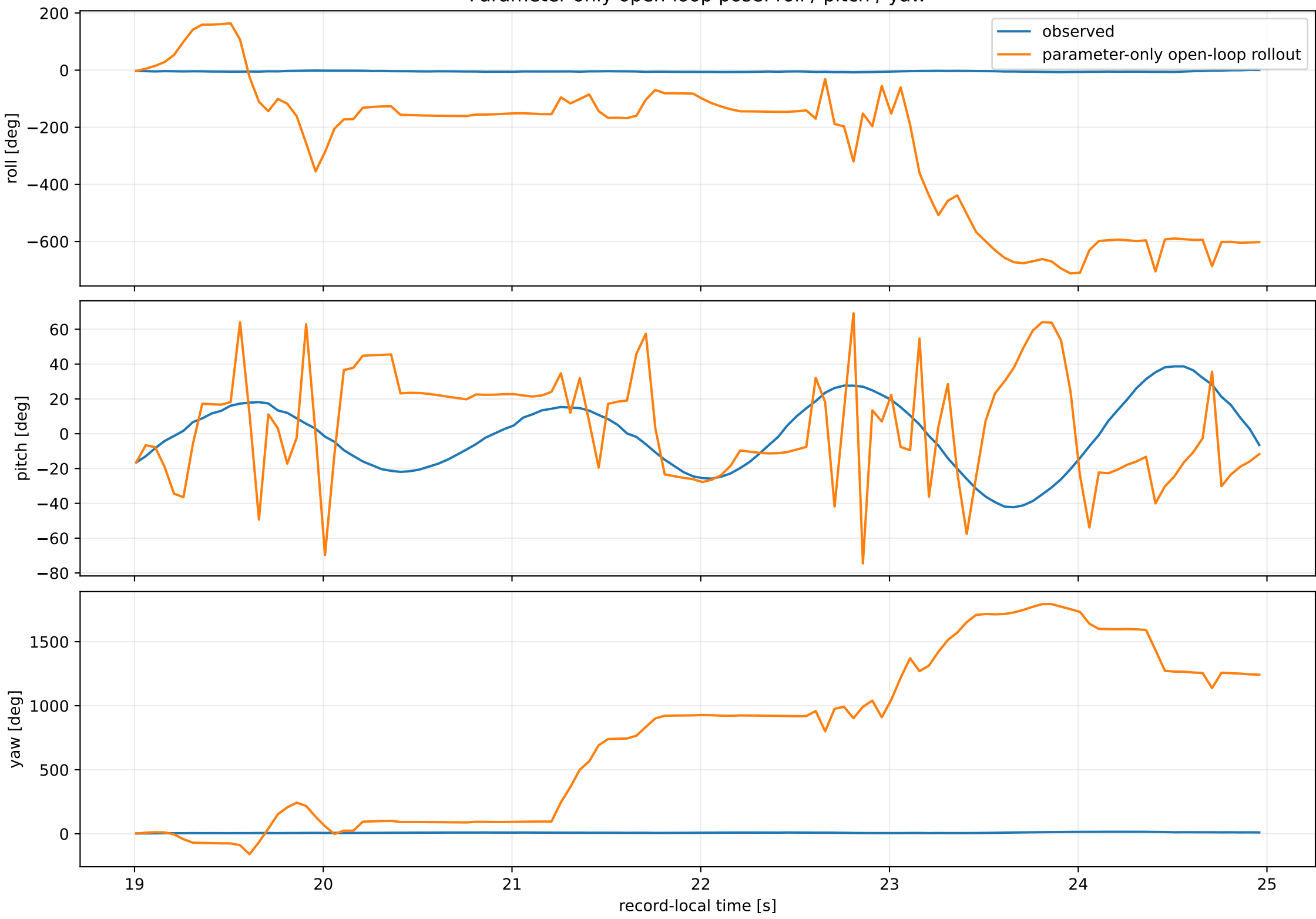
Fine replay error after the 14-D change — lower is better



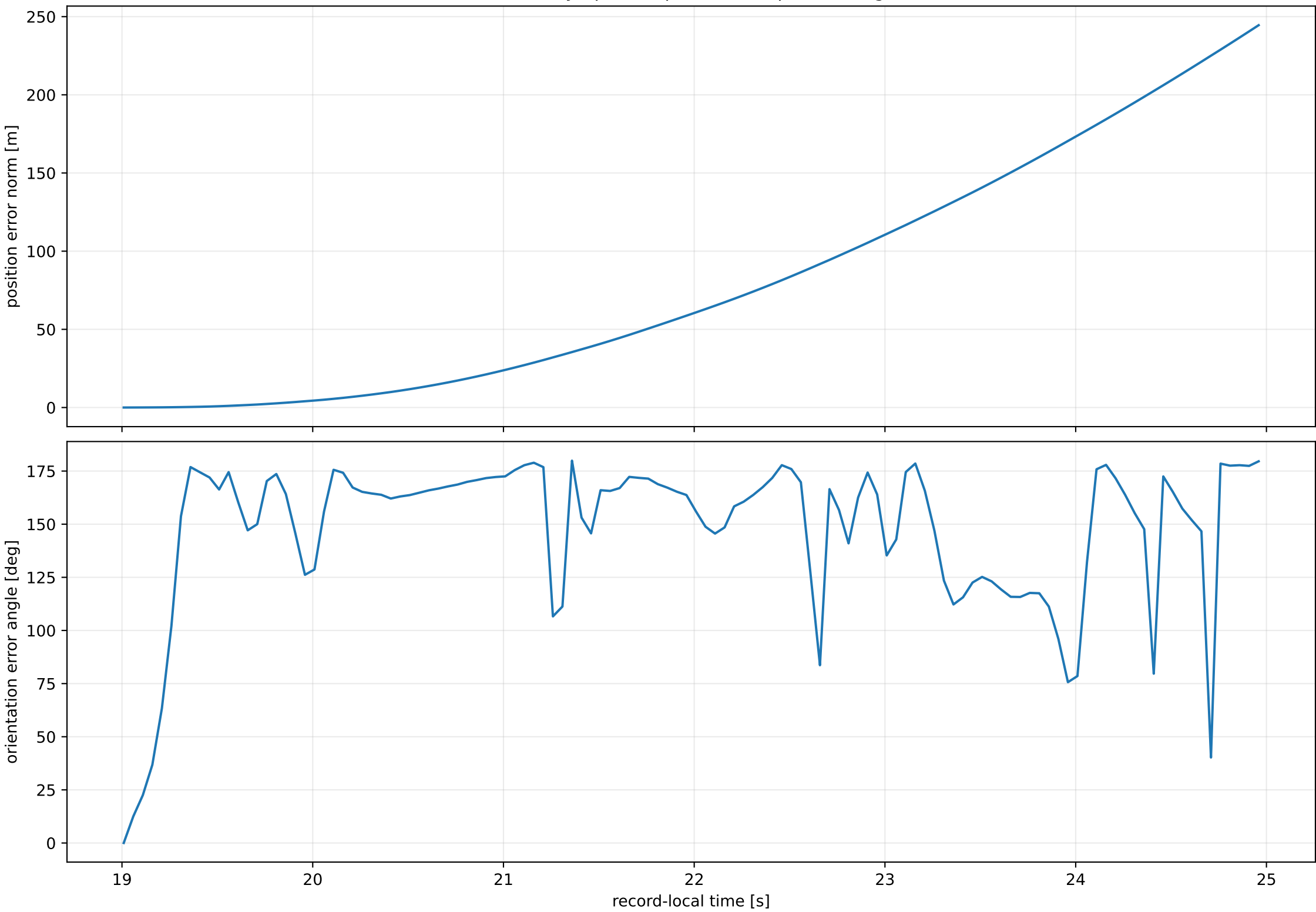
Parameter-only open-loop pose: position



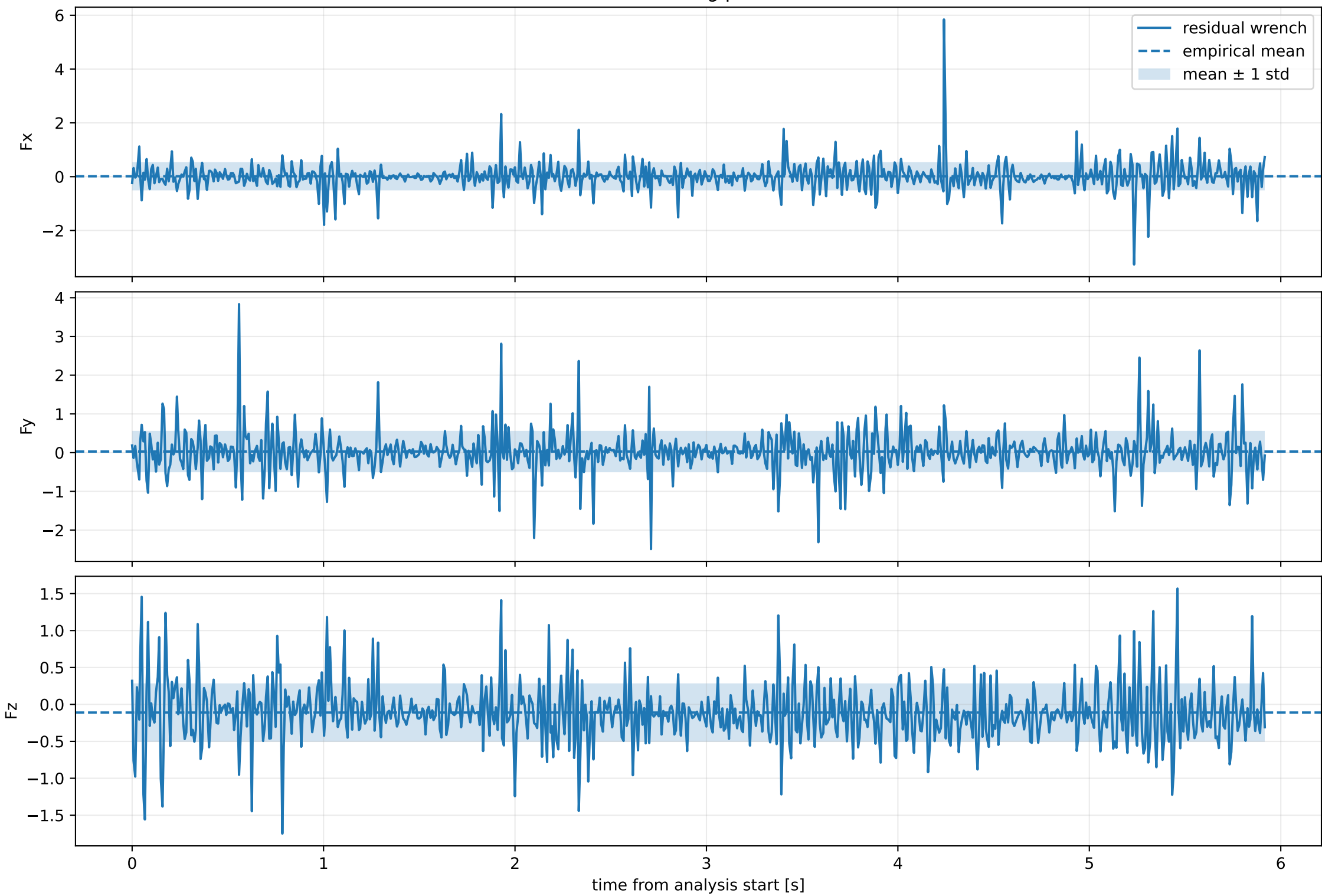
Parameter-only open-loop pose: roll / pitch / yaw



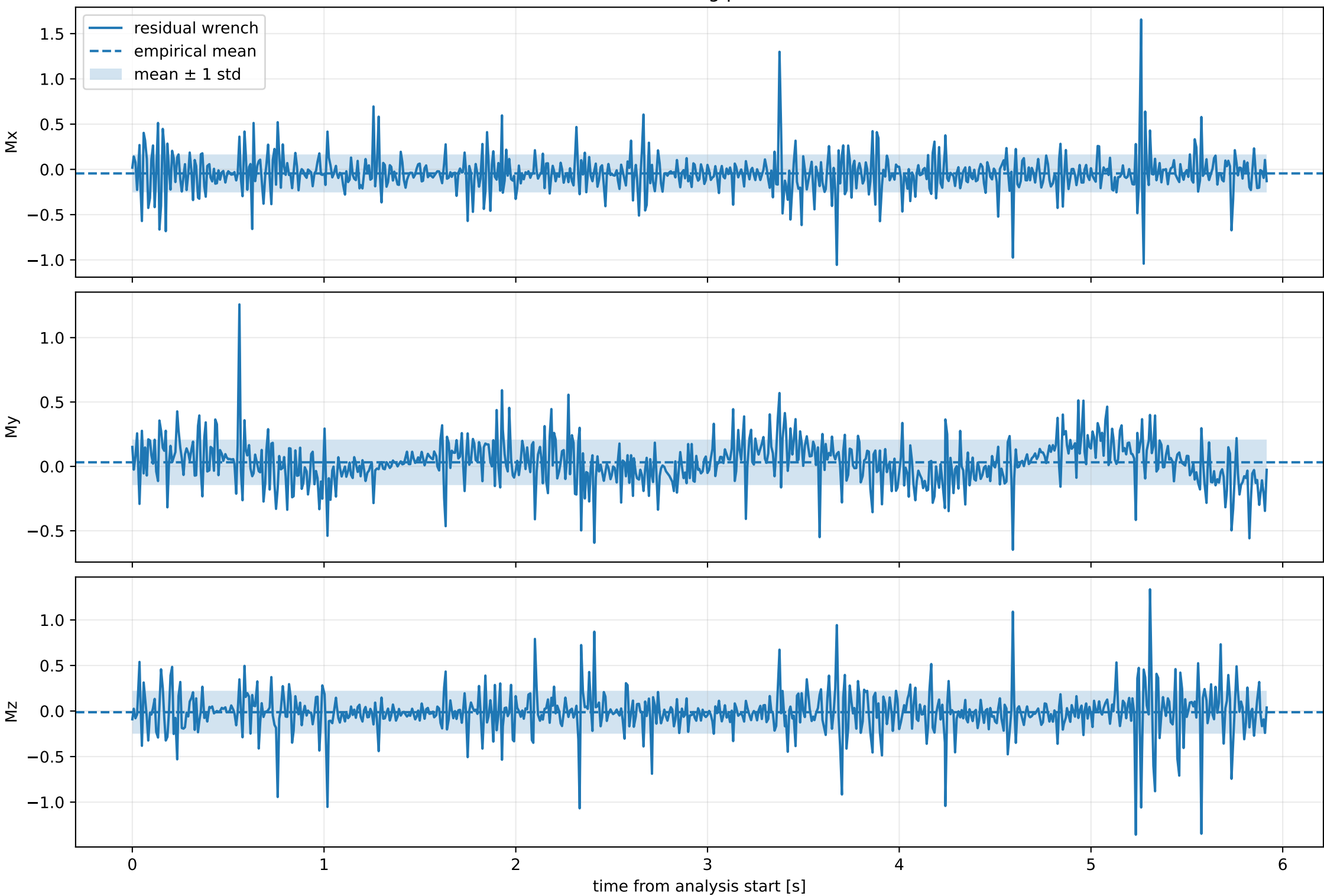
Parameter-only open-loop rollout (separate diagnostic)



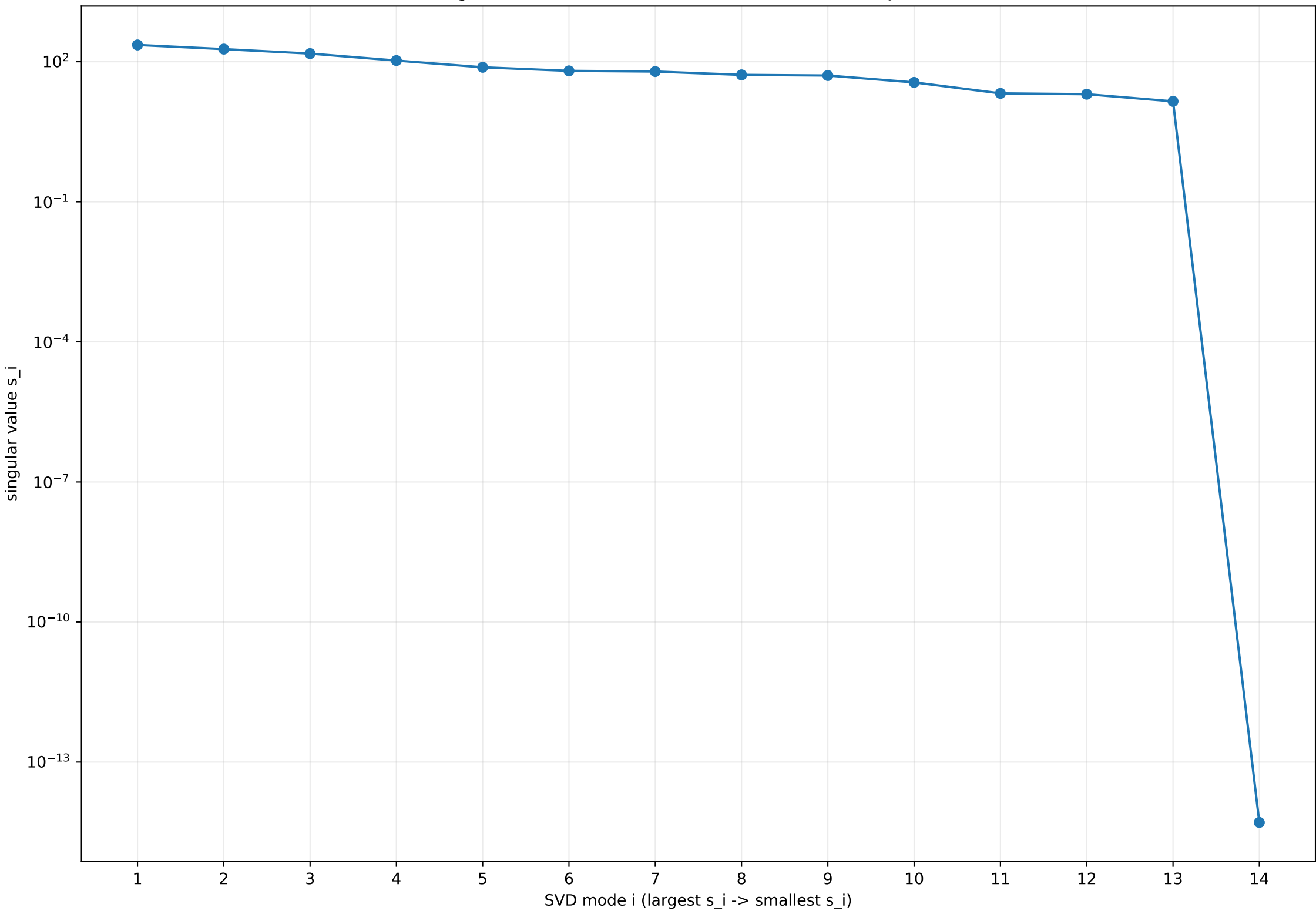
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



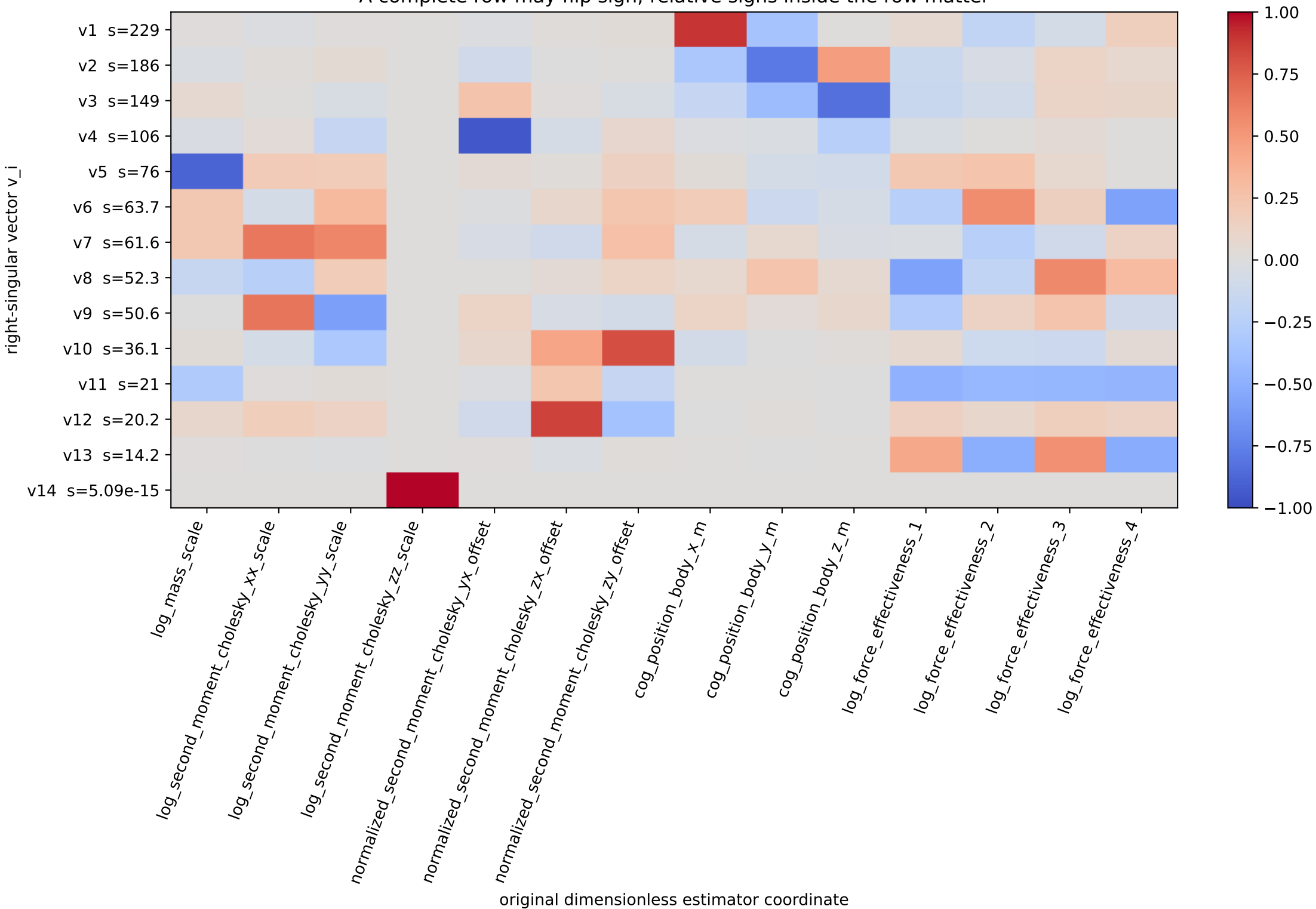
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

1.68629575601

inertia J [kg m²]

[0.0085401773 -0.0013739775 -1.0494247e-05]
[-0.0013739775 0.0059920698 -0.0031166978]
[-1.0494247e-05 -0.0031166978 0.011625927]

principal moments [kg m²]

[0.0042361342, 0.0088429528, 0.013079087]

CoG position in body frame [m]

[-0.029883798, -0.035430487, 0.053668558]

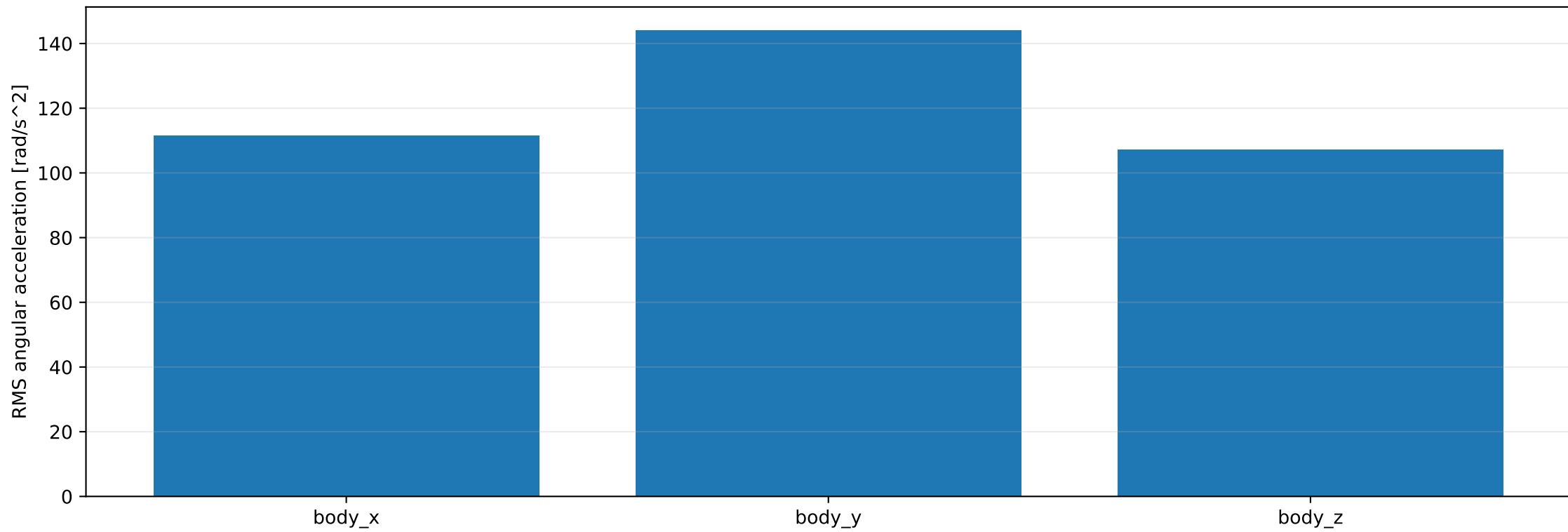
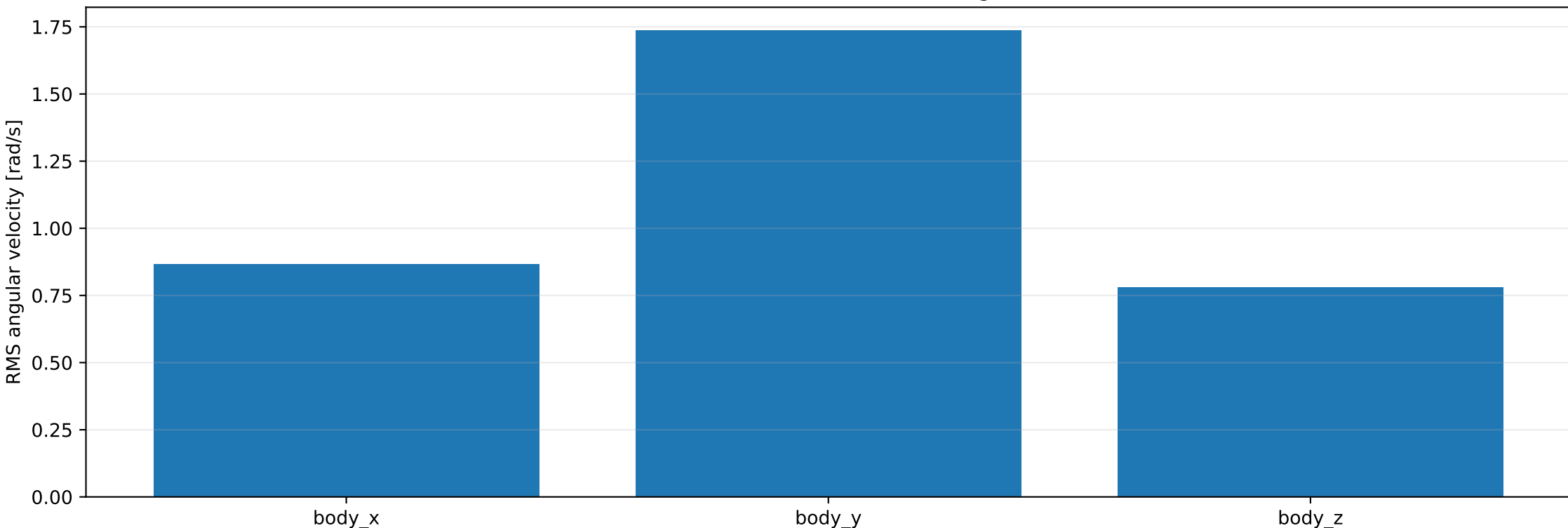
rotor force effectiveness

[0.8692798, 0.71479616, 0.58155249, 0.69146011]

selected delay [s]

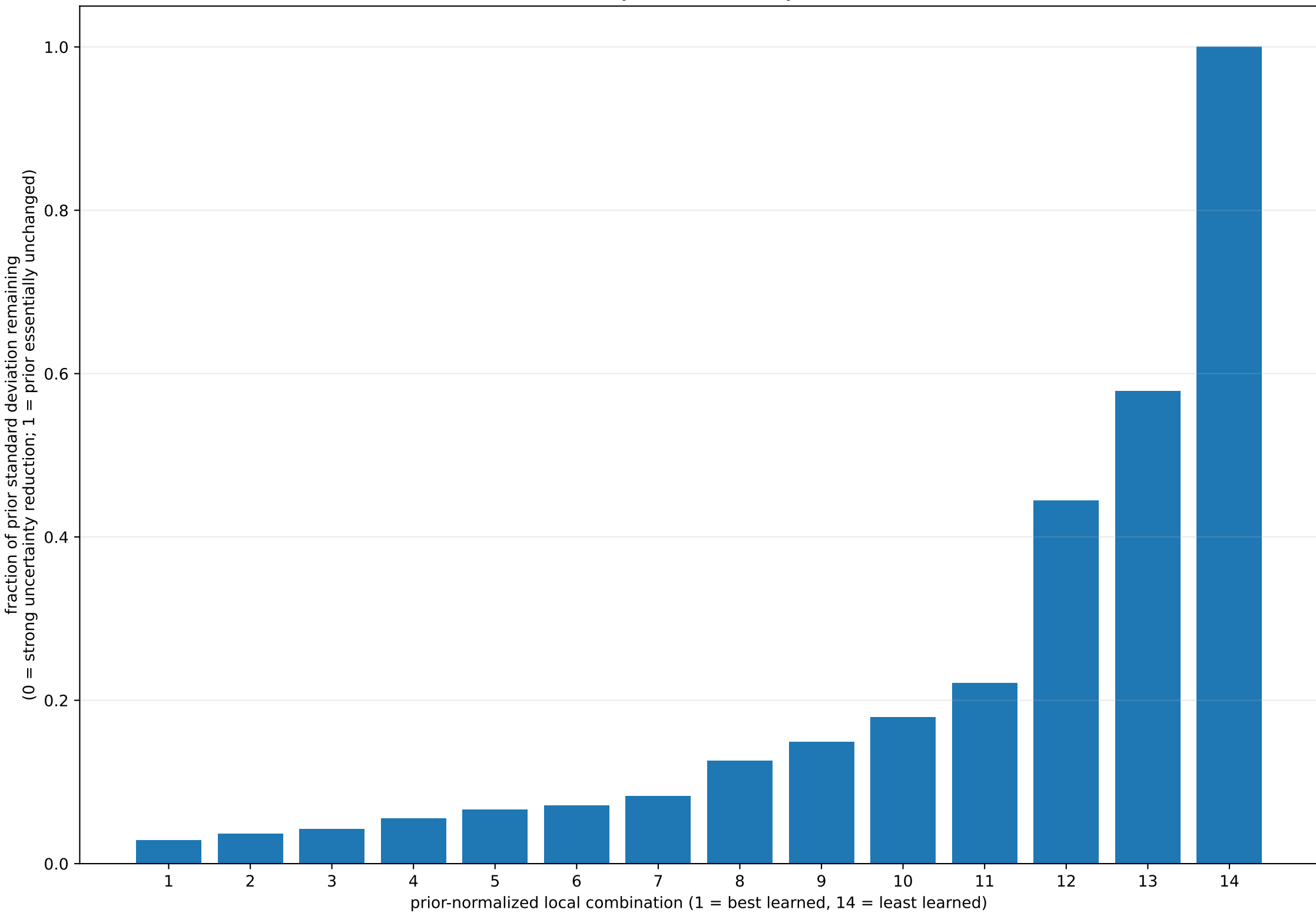
0.032833360527

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

